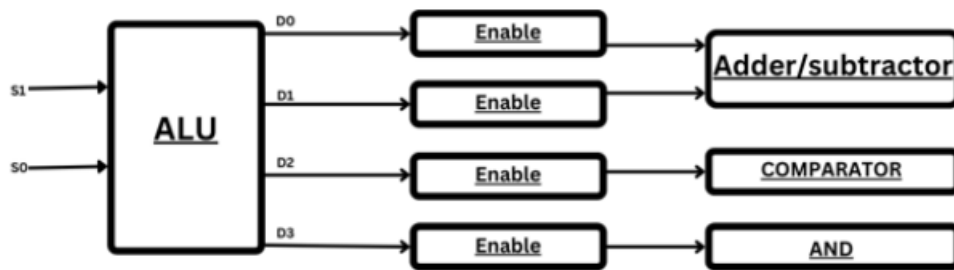


VLSI Project – 4 Bit ALU

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Design of a 4-bit ALU

A 4-bit ALU is an integrated circuit (IC) which can perform operations like addition, subtraction, comparison and ANDing of two 4-bit binary numbers. The design of the 4-bit ALU is shown below:

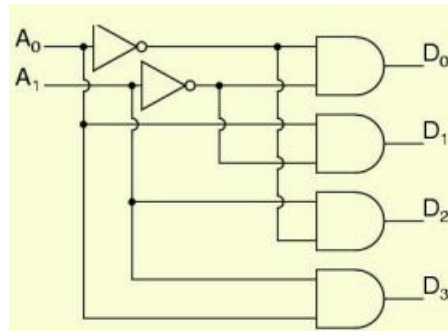


The ALU consists of the following blocks:

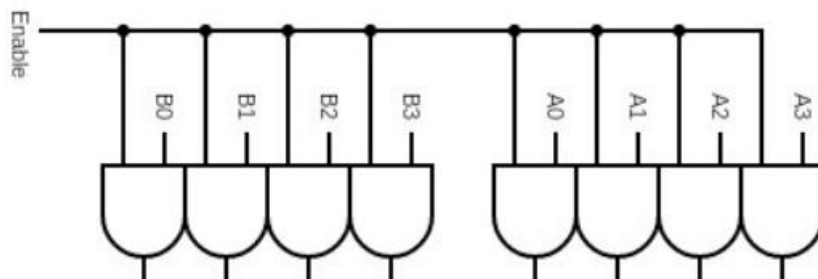
1. Decoder – It is a 2:4 decoder which has 2 inputs – S0 and S1, and gives 4 outputs. Different combinations of S0 and S1 are used to perform different operations as shown below:

S1	S0	Operation
0	0	Addition
0	1	Subtraction
1	0	Comparison
1	1	AND

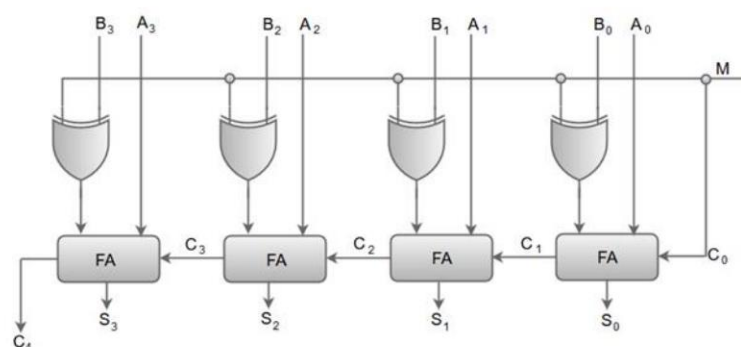
The design of a decoder is shown below:



2. Enable – The enable block is responsible for giving inputs to the operation blocks based on the input received from the decoder. If the input select lines are given for a particular block, then inputs A and B should be given only to that block. Rest of the blocks should be given input as all 0s. This is done by ANDing the output of the decoder block with the input bits – A0, A1, A2, A3, B0, B1, B2 and B3. The design of the enable block is shown below:

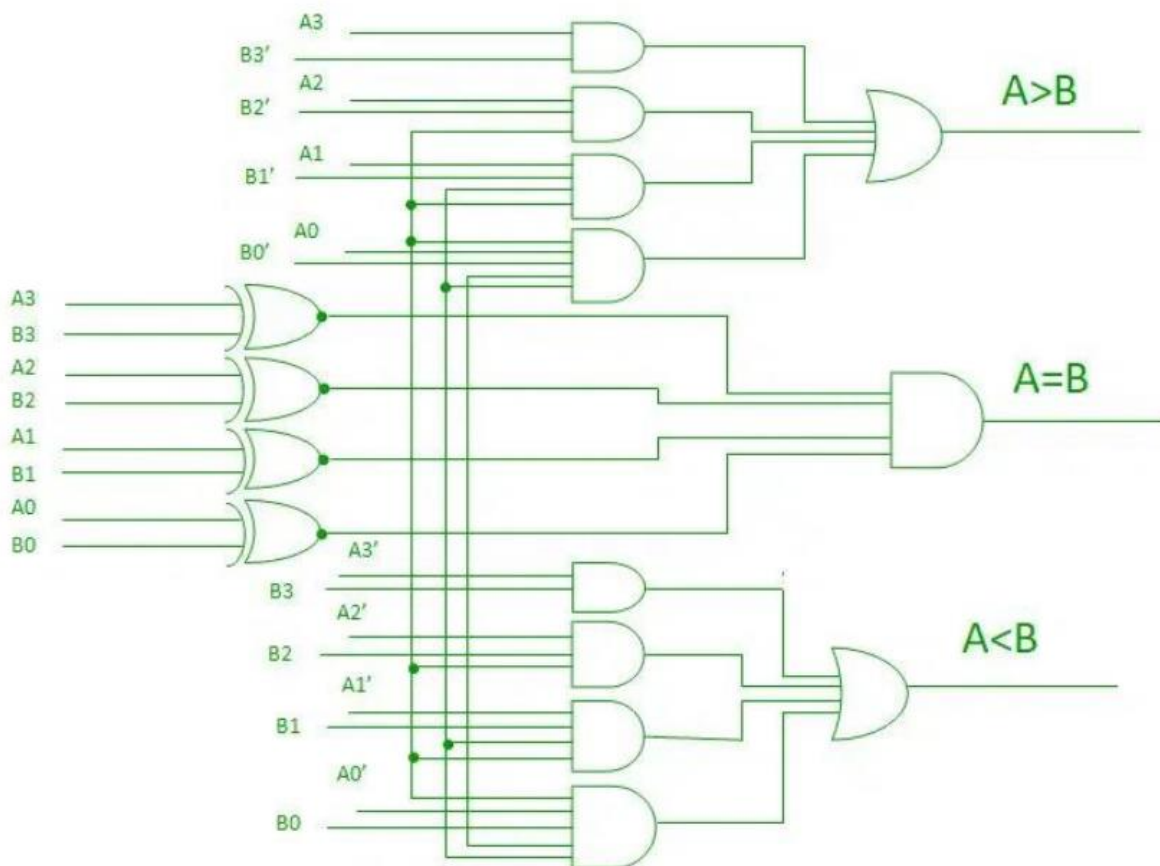


3. Adder – Subtractor – This block is used to add or subtract the 4-bit inputs A and B and give the corresponding 4-bit sum and a carry bit. Addition is carried out when the input is 00 and subtraction is carried out when the input is 01. The circuit diagram of the adder -subtractor block is shown below:

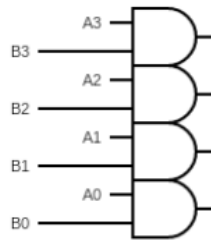


Here, the bit M decides whether addition or subtraction will take place. Therefore, the bit S0 is given as the input to the bit M.

4. Comparator – This block is used to compare the 2 inputs A and B and give the output that which of them is greater or whether both numbers are equal. Therefore, it has 3 outputs – GTA, GTB and EQ. The bit GTA is 1 when $A > B$, the bit GTB is 1 when $B > A$ and the bit EQ is 1 when $A = B$. The design of the comparator block is shown below:



5. AND – This block performs the binary AND between the bits of the two inputs and gives the corresponding output bits. The design of the AND block is shown below:



Implementation in Verilog

The Verilog code for the ALU consisted of an ALU testbench and sub-modules. A testbench is a Verilog code which is responsible for integrating different parts of a circuit and then giving inputs and getting outputs from the circuit. A sub-module is a module which can be used by calling in some other larger module. Multiple sub-modules can be combined to form a single larger module. Therefore, in my implementation of an ALU, I made the following sub-modules:

1. Decoder module
2. Input enable module
3. 1-bit adder-subtractor module
4. Comparator module
5. AND block module

Finally, in the ALU testbench, I called these modules and combined them in order to form a full ALU circuit. This included calling 1 decoder module, 3 input enable modules, 4 adder-subtractor modules, 1 comparator and 1 AND block module.

Implementation in NGSpice

My implementation of an ALU in NGSpice consisted of a single code. In this code I made multiple sub-circuits of individual gates and blocks and then used them to form the final ALU circuit. A sub-circuit is, as the name suggests, a smaller circuit which can be combined with other sub-circuits in order to form a bigger circuit. I made the following sub-circuits in my code:

1. 2-input AND gate
2. 3-input AND gate
3. 4-input AND gate
4. 5-input AND gate
5. Inverter or not gate
6. 2-input OR gate
7. 2-input XOR gate
8. AND block
9. 1-bit full adder-subtractor block
10. 4-bit adder-subtractor block
11. Comparator block

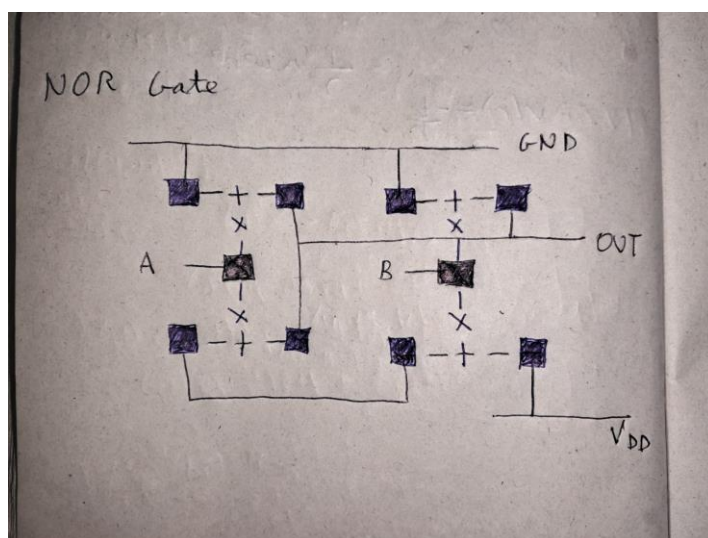
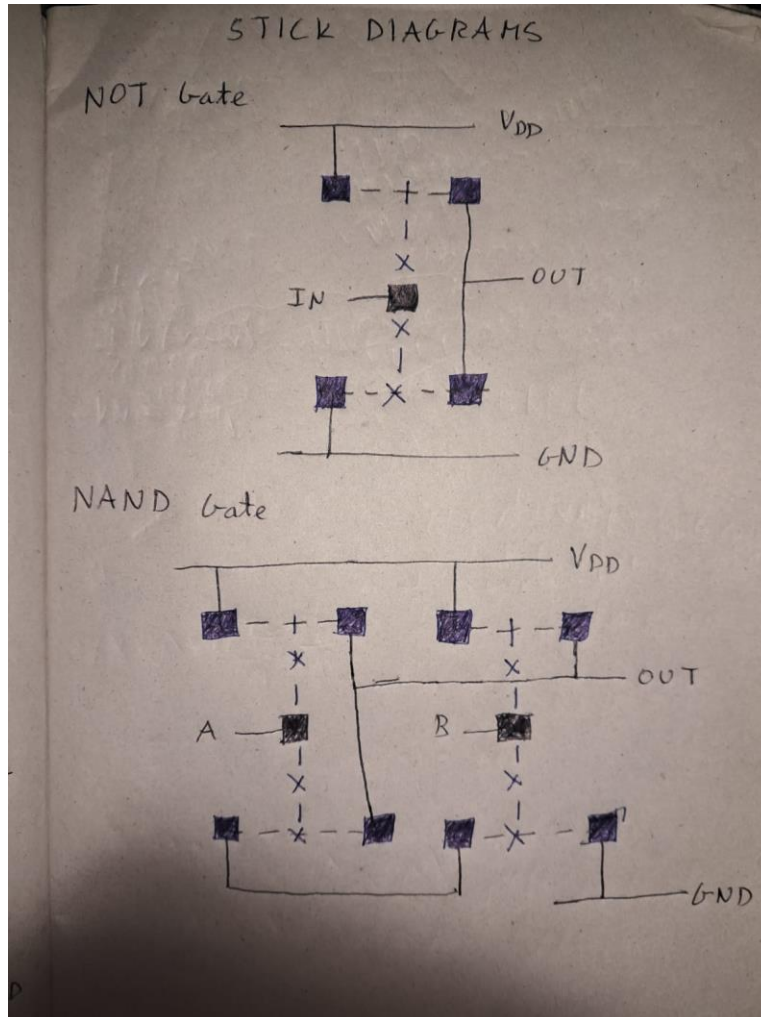
I have all the gates except the XOR gate using individual transistors, that is NMOS and PMOS, and I have made XOR gate and the operation blocks using the gates which act as sub-circuits. Finally, I have combined all the individual blocks in order to form the total circuit of the ALU.

Implementation in Magic

Magic is a software which is used to make layout of circuits. A layout is a design of a circuit which shows how various components are made from p-diffusion, n-diffusion, polysilicon and n-well and then connected using metals such as metal1, metal2, metal3, etc. First, I made individual basic gates such as 2-input NAND gate, not gate or inverter, 3-input NAND gate, 4-input NAND gate, 2-input NOR gate, etc using the CMOS design consisting of NMOS and PMOS. The NMOS and PMOS are made using materials like p-diffusion, n-diffusion, polysilicon and n-well. After making the basic gates, I make those gates into sub-cells. A sub-cell is a smaller sub-part of a larger layout. Multiple sub-cells can be used to form a larger layout. Thus, I use the sub-cells of the basic gates in order to make other gates like 2-input AND, 2-input OR, XOR gate, XNOR gate, etc. Next, I use all these sub-cells of

gates to form individual operation blocks. Finally, I combine all the individual operation blocks in order to form my overall ALU circuit.

Stick Diagram of Basic Gates



Challenges and their Solutions

1. **Challenge** – The decoder has 4 outputs and there are 4 enable blocks but there are only 3 operation blocks.

Solution – In order to solve this problem, we take a single enable block instead of 2 enable blocks for the adder-subtractor block. The enable for the adder – subtractor block is taken to be S_1' . This is because when S_1 is 0, either addition or subtraction takes place but when S_1 is 1, neither of them takes place.

2. **Challenge** – The adder-subtractor block has only 1 enable block connected to it but it performs 2 operations, that is, addition and subtraction.

Solution – The adder-subtractor has a single set of input but whether addition or subtraction is performed is decided based on the value of the bit M . The second bit, that is, S_0 is given as the input M . When $S_0=0$, addition takes place and when $S_0 = 1$, subtraction takes place.

3. **Challenge** - When the input S_0 and S_1 is given for blocks other than that for the comparator blocks, the enable for the comparator block is 0. So, the inputs to the comparator block are also all 0s. As $A = B = 0000$, both A and B are equal and hence the EQ bit is 1. But actually, the comparator block is not enabled so all three outputs of the comparator block should be 0.

Solution- In order to solve this issue, I AND the output bit EQ with S_0' and S_1 . Therefore, the bit EQ can show the output 1 only when $S_0 = 0$ and $S_1 = 1$, that is, when the comparator block is enabled. For other inputs of S_0 and S_1 as $S_0' \text{ AND } S_1$ will be 0, the output EQ will always be 0.

Delay Analysis

1. Addition block:

(a) Delay between A and sum[0]

trise	=	1.526000e-08	targ=	6.526000e-08	trig=	5.000000e-08
tfall	=	1.350449e-08	targ=	2.016350e-05	trig=	2.015000e-05
tpd	=	1.43822e-08				

(b) Delay between A and sum[1]

trise	=	1.526000e-08	targ=	6.526000e-08	trig=	5.000000e-08
tfall	=	1.350449e-08	targ=	2.016350e-05	trig=	2.015000e-05
tpd	=	1.43822e-08				

(c) Delay between A and sum[2]

trise	=	1.526000e-08	targ=	6.526000e-08	trig=	5.000000e-08
tfall	=	1.350449e-08	targ=	2.016350e-05	trig=	2.015000e-05
tpd	=	1.43822e-08				

(d) Delay between A and sum[3]

trise	=	1.526000e-08	targ=	6.526000e-08	trig=	5.000000e-08
tfall	=	1.350449e-08	targ=	2.016350e-05	trig=	2.015000e-05
tpd	=	1.43822e-08				

(e) Delay between B and sum[0]

trise	=	1.526283e-08	targ=	6.526283e-08	trig=	5.000000e-08
tfall	=	1.363908e-08	targ=	2.016364e-05	trig=	2.015000e-05
tpd	=	1.44510e-08				

(f) Delay between B and sum[1]

trise	=	1.526283e-08	targ=	6.526283e-08	trig=	5.000000e-08
tfall	=	1.363908e-08	targ=	2.016364e-05	trig=	2.015000e-05
tpd	=	1.44510e-08				

(g) Delay between B and sum[2]

trise	=	1.526283e-08	targ=	6.526283e-08	trig=	5.000000e-08
tfall	=	1.363908e-08	targ=	2.016364e-05	trig=	2.015000e-05
tpd	=	1.44510e-08				

(h) Delay between B and sum[3]

trise	=	1.526283e-08	targ=	6.526283e-08	trig=	5.000000e-08
tfall	=	1.363908e-08	targ=	2.016364e-05	trig=	2.015000e-05
tpd	=	1.44510e-08				

2. Subtractor block:

(a) Delay between A and sum[0]

trise	=	1.522072e-08	targ=	6.522072e-08	trig=	5.000000e-08
tfall	=	1.008152e-08	targ=	2.016008e-05	trig=	2.015000e-05
tpd	=	1.26511e-08				

(b) Delay between A and sum[1]

trise	=	1.522072e-08	targ=	6.522072e-08	trig=	5.000000e-08
tfall	=	1.008152e-08	targ=	2.016008e-05	trig=	2.015000e-05
tpd	=	1.26511e-08				

(c) Delay between A and sum[2]

trise	=	1.522072e-08	targ=	6.522072e-08	trig=	5.000000e-08
tfall	=	1.008152e-08	targ=	2.016008e-05	trig=	2.015000e-05
tpd	=	1.26511e-08				

(d) Delay between A and sum[3]

trise	=	1.522072e-08	targ=	6.522072e-08	trig=	5.000000e-08
tfall	=	1.008152e-08	targ=	2.016008e-05	trig=	2.015000e-05
tpd	=	1.26511e-08				

(e) Delay between B and sum[0]

trise	=	1.518883e-08	targ=	6.518883e-08	trig=	5.000000e-08
tfall	=	1.338702e-08	targ=	2.016339e-05	trig=	2.015000e-05
tpd	=	1.42879e-08				

(f) Delay between B and sum[1]

trise	=	6.010821e-05	targ=	6.015821e-05	trig=	5.000000e-08
tfall	=	4.001006e-05	targ=	6.016006e-05	trig=	2.015000e-05
tpd	=	5.00591e-05				

(g) Delay between B and sum[2]

trise	=	6.010821e-05	targ=	6.015821e-05	trig=	5.000000e-08
tfall	=	4.001136e-05	targ=	6.016136e-05	trig=	2.015000e-05
tpd	=	5.00598e-05				

(h) Delay between B and sum[3]

trise	=	6.010821e-05	targ=	6.015821e-05	trig=	5.000000e-08
tfall	=	4.001353e-05	targ=	6.016353e-05	trig=	2.015000e-05
tpd	=	5.00609e-05				

3. Comparator:

(a) Delay between A[0] and EQ

trise	=	1.519019e-08	targ=	6.519019e-08	trig=	5.000000e-08
tfall	=	1.014268e-08	targ=	2.016014e-05	trig=	2.015000e-05
tpd	=	1.26664e-08				

(b) Delay between A[1] and EQ

trise	=	1.519019e-08	targ=	6.519019e-08	trig=	5.000000e-08
tfall	=	1.014268e-08	targ=	2.016014e-05	trig=	2.015000e-05
tpd	=	1.26664e-08				

(c) Delay between A[2] and EQ

trise	=	1.519019e-08	targ=	6.519019e-08	trig=	5.000000e-08
tfall	=	1.014268e-08	targ=	2.016014e-05	trig=	2.015000e-05
tpd	=	1.26664e-08				

(d) Delay between A[3] and EQ

trise	=	1.519019e-08	targ=	6.519019e-08	trig=	5.000000e-08
tfall	=	1.014268e-08	targ=	2.016014e-05	trig=	2.015000e-05
tpd	=	1.26664e-08				

(e) Delay between B[0] and EQ

trise	=	1.515955e-08	targ=	6.515955e-08	trig=	5.000000e-08
tfall	=	1.345196e-08	targ=	2.016345e-05	trig=	2.015000e-05
tpd	=	1.43058e-08				

(f) Delay between B[1] and EQ

trise	=	1.515955e-08	targ=	6.515955e-08	trig=	5.000000e-08
tfall	=	1.345196e-08	targ=	2.016345e-05	trig=	2.015000e-05
tpd	=	1.43058e-08				

(g) Delay between B[2] and EQ

trise	=	1.515955e-08	targ=	6.515955e-08	trig=	5.000000e-08
tfall	=	1.345196e-08	targ=	2.016345e-05	trig=	2.015000e-05
tpd	=	1.43058e-08				

(h) Delay between B[3] and EQ

trise	=	1.515955e-08	targ=	6.515955e-08	trig=	5.000000e-08
tfall	=	1.345196e-08	targ=	2.016345e-05	trig=	2.015000e-05
tpd	=	1.43058e-08				

(i) Delay between B and GTA

trise	=	1.516581e-08	targ=	6.516581e-08	trig=	5.000000e-08
tfall	=	7.179861e-09	targ=	2.015718e-05	trig=	2.015000e-05
tpd	=	1.11728e-08				

(j) Delay between A and GTA

trise	=	1.525635e-08	targ=	6.525635e-08	trig=	5.000000e-08
tfall	=	1.007185e-08	targ=	2.016007e-05	trig=	2.015000e-05
tpd	=	1.26641e-08				

(k) Delay between B and GTB

trise	=	1.521697e-08	targ=	6.521697e-08	trig=	5.000000e-08
tfall	=	1.336803e-08	targ=	2.016337e-05	trig=	2.015000e-05
tpd	=	1.42925e-08				

(l) Delay between A and GTB

trise	=	1.132653e-08	targ=	6.132653e-08	trig=	5.000000e-08
tfall	=	1.326596e-08	targ=	2.016327e-05	trig=	2.015000e-05
tpd	=	1.22962e-08				

4. AND block:

(a) Delay between A[3] and and[3]

trise	=	1.523036e-08	targ=	6.523036e-08	trig=	5.000000e-08
tfall	=	1.588035e-08	targ=	2.016588e-05	trig=	2.015000e-05
tpd	=	1.55554e-08				

(b) Delay between A[2] and and[2]

trise	=	1.523036e-08	targ=	6.523036e-08	trig=	5.000000e-08
tfall	=	1.588035e-08	targ=	2.016588e-05	trig=	2.015000e-05
tpd	=	1.55554e-08				

(c) Delay between A[1] and and[1]

trise	=	1.523036e-08	targ=	6.523036e-08	trig=	5.000000e-08
tfall	=	1.588035e-08	targ=	2.016588e-05	trig=	2.015000e-05
tpd	=	1.55554e-08				

(d) Delay between A[0] and and[0]

trise	=	1.523036e-08	targ=	6.523036e-08	trig=	5.000000e-08
tfall	=	1.588035e-08	targ=	2.016588e-05	trig=	2.015000e-05
tpd	=	1.55554e-08				

(e) Delay between B[3] and and[3]

trise	=	1.523034e-08	targ=	6.523034e-08	trig=	5.000000e-08
tfall	=	9.971517e-09	targ=	2.015997e-05	trig=	2.015000e-05
tpd	=	1.26009e-08				

(f) Delay between B[2] and and[2]

trise	=	1.523034e-08	targ=	6.523034e-08	trig=	5.000000e-08
tfall	=	9.971517e-09	targ=	2.015997e-05	trig=	2.015000e-05
tpd	=	1.26009e-08				

(g) Delay between B[1] and and[1]

```
trise      = 1.523034e-08  targ= 6.523034e-08  trig= 5.000000e-08
tfall      = 9.971517e-09  targ= 2.015997e-05  trig= 2.015000e-05
tpd        = 1.26009e-08
```

(h) Delay between B[0] and and[0]

```
trise      = 1.523034e-08  targ= 6.523034e-08  trig= 5.000000e-08
tfall      = 9.971517e-09  targ= 2.015997e-05  trig= 2.015000e-05
tpd        = 1.26009e-08
```

Conclusion based on Delay Analysis

Block	Maximum Delay
Adder	1.44510e-08
Subtractor	5.00591e-05
Comparator	1.43058e-08
AND	1.26009e-08

Based on the delay analysis of all the blocks, the maximum delay is found in the case of the subtractor. The maximum delay is 5.00591e-05 seconds, that is, close to 50 micro seconds. Therefore, the path from the inputs A and B to the output sum[3] can be considered to be the critical path of the 4-bit ALU. It will take the longest time for computation.